

A sign-change bound for strictly increasing comparison columns

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For a nonzero real vector h on the ordered set $[K]$, write $\text{var}(h)$ for the number of sign changes after zero entries are deleted. Expectation below may use any strictly positive weights on $[K]$.

Lemma 1 (Products detect few sign changes). *Let $f_1, \dots, f_d$ be strictly increasing functions on $[K]$. If a nonzero vector h satisfies*

$$\mathbb{E} \left[h \prod_{j \in S} f_j \right] = 0 \quad \text{for every } S \subseteq [d],$$

then $\text{var}(h) \geq d + 1$.

Proof. Suppose h has $r \leq d$ sign changes. Assign a distinct f_j to each sign change. If consecutive nonzero entries $h(q), h(q')$ have opposite signs, choose the threshold for its assigned function strictly between $f_j(q)$ and $f_j(q')$. The product of these r factors $(f_j - \theta_j)$ changes sign at exactly the same places as h on the support of h , and is nonzero there. Changing its overall sign if necessary gives a function P such that $hP > 0$ throughout that support. But P expands into squarefree products of the f_j , so $\mathbb{E}[hP] = 0$, a contradiction. $\square$

Lemma 2 (Variation of a product). *For nonzero vectors h, g on $[K]$ whose pointwise product is nonzero,*

$$\text{var}(hg) \leq 2K - 2 - \text{var}(h) - \text{var}(g).$$

Proof. First suppose neither vector has a zero. An edge on which the product changes sign is an edge on which exactly one factor changes sign. Counting the four possible types of edge gives $\text{var}(hg) \leq 2(K - 1) - \text{var}(h) - \text{var}(g)$.

For the general case, let z_h, z_g be their numbers of zeros and let z_0 count their common zeros. Restrict both vectors to their common nonzero support, of size $N = K - z_h - z_g + z_0$. Deleting a nonzero entry reduces sign variation by at most two. The restricted variations are therefore at least $\text{var}(h) - 2(z_g - z_0)$ and $\text{var}(g) - 2(z_h - z_0)$. Applying the zero-free bound on the common support gives

$$\text{var}(hg) \leq 2K - 2 - \text{var}(h) - \text{var}(g) - 2z_0,$$

which suffices. $\square$

Theorem 3. *Let $m \geq 4$. Suppose $p_1, \dots, p_m$ are strictly increasing functions on $[K]$, and their mixed squarefree moments depend only on the number of factors:*

$$\mathbb{E} \prod_{j \in S} p_j = \mu_{|S|} \quad (S \subseteq [m]).$$

If $K < (3m - 3)/2$, then at least $m - 1$ of the functions are identical.

Proof. For distinct indices i, j , the difference $h = p_i - p_j$ is orthogonal to all squarefree products of the remaining $m - 2$ functions. Lemma 1 shows that either $h = 0$ or $\text{var}(h) \geq m - 1$. In particular every nonzero difference has at least m nonzero entries.

Suppose there are two disjoint pairs of indices with nonzero differences, say $h = p_i - p_j$ and $g = p_k - p_l$. Since $K < (3m - 3)/2 < 2m$, their supports intersect and $hg \neq 0$. Expanding the differences shows that hg is orthogonal to all squarefree products of the remaining $m - 4$ functions. Hence Lemma 1 gives $\text{var}(hg) \geq m - 3$. But Lemma 2 gives

$$\text{var}(hg) \leq 2K - 2 - 2(m - 1) = 2(K - m) < m - 3,$$

a contradiction.

Thus the graph joining indices with different functions has no matching of size two. It is a complete multipartite graph, its parts being the classes of identical functions. For $m \geq 4$, such a graph can have matching number at most one only if one class has size at least $m - 1$. $\square$

Corollary 4 (Rational uniform comparison tensors). *Suppose additionally that all $p_j(q)$ are rational, the K atoms have equal weight, and $\mu_s = 1/(s + 1)$. For $m \geq 4$,*

$$K \geq \left\lceil \frac{3m - 3}{2} \right\rceil.$$

Thus a permutation-fair collection of $n \geq 5$ finite dice has this bound, with $m = n - 1$, on a distinguished die whenever every other die has at least one face strictly between each consecutive pair of its faces.

Proof. Otherwise Theorem 3 supplies a common rational column occurring at least $m - 1$ times. Its K values are an equal-weight uniform-interval quadrature rule of degree $m - 1$. For $m = 4, 5, 6, 7, 8$, the odd-prime obstruction in `odd_prime_quadrature_obstruction.tex` gives, respectively, $K \geq 6, 10, 10, 14, 14$, contradicting $K < (3m - 3)/2$. For $m \geq 9$, the bound $K \geq 2^{\lfloor (m-1)/2 \rfloor}$ from `./rational_designs/two_adic_lower_bound.tex` is already larger than $(3m - 3)/2$. $\square$

Lemma 5 (A sparse-plateau version of Lemma 1). *Let $f_1, \dots, f_d$ be nondecreasing. Suppose each f_j is flat on at most b adjacent pairs of $[K]$, and at most F of the functions are flat on any one adjacent pair. If $b + F \leq d$, then the conclusion of Lemma 1 still holds.*

Proof. Suppose a nonzero orthogonal h has $r \leq d$ sign changes. Form the bipartite graph from these r cuts to the d functions: a function is adjacent to a cut between consecutive nonzero entries $h(q), h(q')$ when $f_j(q) < f_j(q')$. Each cut has at least $d - F \geq b$ neighbors. Each function fails to be adjacent to at most b cuts, because the cut intervals are disjoint and flatness across one consumes at least one of its flat adjacent pairs. A set of at most b cuts therefore has at least as many neighbors, and a set of more than b cuts has all d functions as neighbors. Hall's theorem gives distinct functions for all cuts. The threshold-product proof of Lemma 1 now applies. $\square$

Corollary 6 (Sparse plateaus at $K = m$). *Suppose the functions in Theorem 3 are only nondecreasing, and $K = m$. Define b as the largest number of flat adjacent pairs of one column, and F as the largest number of columns flat on one adjacent pair. If $b + F \leq m - 2$, then at least $m - 1$ columns are identical. In the rational uniform-interval setting, no such tensor exists for $m \geq 4$.*

Proof. For any nonzero difference $h = p_i - p_j$, Lemma 5 applies to the other $m - 2$ columns. Thus h changes sign at least $m - 1$ times; as there are m atoms, it is nonzero everywhere and strictly alternating. Two differences from disjoint pairs would then have a product with a

constant nonzero sign. This contradicts their orthogonality to each other. The same complete multipartite graph argument as before gives $m - 1$ identical columns.

These common values give a rational equal-weight degree- $(m - 1)$ rule with m nodes. For $m = 4$ the degree-three bound $K \geq 6$ from `odd_prime_quadrature_obstruction.tex` excludes it. For $m \geq 5$, `optimized_two_adic_filter.tex` gives $K \geq 2^{2\lfloor(m-1)/2\rfloor} > m$. $\square$

The proof of Theorem 3 itself also extends to nondecreasing columns whenever $b + F \leq m - 4$: apply Lemma 5 both to single differences and to products of two disjoint differences. Thus the same threshold $(3m - 3)/2$ and its rational corollary remain valid under that sparse-plateau condition.

Corollary 7 (One flat edge per column). *In the rational uniform-interval setting with $K = m \geq 5$, some column must be flat on at least two adjacent pairs.*

Proof. Suppose every column is flat on at most one edge. If $F \leq m - 3$, Corollary 6 already gives a contradiction. Thus some edge e has at least $m - 2$ flat columns. There is at most one such edge, because two sets of $m - 2$ columns would intersect for $m \geq 5$.

If exactly two columns p_a, p_b increase at e , choose distinct flat columns p_c, p_d . Both differences $p_a - p_c$ and $p_b - p_d$ are nonzero. For each difference, the other $m - 2$ columns include one increasing column at e . At every other edge at most two columns can be flat, so again the unused $m - 2$ columns are not all flat. In each of the two cut-function graphs, every cut therefore has a neighbor, and every function misses at most one cut. The Hall argument in Lemma 5, with $b = 1$, applies. Both differences must alternate on all m atoms, contradicting the zero expectation of their product.

If exactly one column increases at e , the same Hall argument applies to the difference of any two flat columns. Such a difference is flat across e , so it cannot alternate on all m atoms. All $m - 1$ flat columns are therefore identical, which gives the excluded rational degree- $(m - 1)$ quadrature.

Finally, if every column is flat at e , merge its two adjacent atoms. There are now at most $m - 1$ ordered atoms with positive weights. The structural proof in `monotone_tensor.tex`, Theorem 1, works with arbitrary positive atom weights: its Gram argument, interpolation, and the equality case of the monotone covariance inequality use only positivity of those weights. It therefore gives a single column whose scalar moments through degree m are uniform interval moments. Undoing the merge repeats its value and makes this an equal-weight rational rule with m atoms, again impossible by `optimized_two_adic_filter.tex`. $\square$

Scope and the unresolved plateau case. Without a condition such as Lemma 5, strict increase is essential to the proof of Lemma 1: if an assigned function is flat across a sign change, its threshold factor can vanish on that entire plateau. A product of these factors can then vanish on all of the support of h , so there is no positivity contradiction. The theorem as written does not give a universal bound for arbitrary finite dice. Failure of the matching used above would need a separate structural analysis of common plateaus. In particular this note does not establish a quadratic per-die bound. For the unresolved case $m = K = 5$, Corollary 6 shows that any rational counterexample must have $b + F \geq 4$. Corollary 7 further requires $b \geq 2$, so at least one column has at most three distinct values.